

Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race

Abstract

We study evolutionary dynamics in which the payoff earned in the game and the payoff that governs replication are different functions of the same interaction. This separation is the defining feature of an ecosystem of deployed AI agents: an agent acts in a competitive race, but whether its *design* is replicated is decided by a principal who does not play that race and who bears only a fraction λ of the external harm the design causes. We call the difference between the two functionals the selection wedge, embed it in the reduced four-strategy AI race of Fernández Domingos and Han, and analyse the resulting replicator and finite-population dynamics. The wedge collapses to a single control parameter, the effective liability $L = \lambda h$, and every invasion condition is affine in L , which yields closed-form thresholds. Four consequences follow. First, a conditionally unsafe design weakly dominates the unconditionally unsafe one at every L . A pre-deployment filter that scores designs by their behaviour against a safe environment therefore removes only a weakly dominated design, deletes only the attractor of maximal harm, which the surviving pool reproduces, and leaves the long-run unsafe frequency of the same ecosystem unchanged. Second, unconditional and conditional safety are exact payoff twins, so the invasion barrier of the safe face is set by a neutrally drifting composition variable and varies by a factor of 77.7 along that face. Third, long-run harm is *not monotone* in liability. On the window $L \in (4.27, 42.77)$ liability penalises the retaliation that conditional safety relies on, the unsafe coexistence becomes uninvadable, and raising L across the lower edge lifts the basin-averaged unsafe frequency from 0 to 0.32. Fourth, coupling the dynamics to a non-decreasing stock of diffused capability that erodes enforcement by a factor θ produces hysteresis of exact width $1/(1 - \theta)$. The results locate the governance problem at the selection layer rather than at the design layer, and identify liability regimes that are worse than either extreme.

Keywords: evolutionary game theory, replicator dynamics, technological race, bistability, hysteresis, liability, AI safety

1. Introduction

Evolutionary game theory rests on an identity that is rarely stated because it is rarely violated: the payoff an individual earns in the game is the same quantity that governs how often that individual is copied [1, 2, 3]. Fitness is the game payoff. Every standard result — the folk theorem linking rest points to Nash equilibria, the stability of evolutionarily stable strategies, the invasion analysis of finite populations — is a statement about one payoff matrix used twice.

Ecosystems of deployed AI agents break that identity. An agent negotiates, bids, schedules or races on behalf of a principal, and earns a payoff by doing so. Whether the agent’s *design* — its model, its system prompt, its scaffold — is replicated across further deployments is not decided by the agent and is not decided by the payoff the agent earned in the abstract. It is decided by the principal who deployed it, on the basis of what the principal received. The two quantities differ whenever the design imposes costs on third parties that the principal does not bear. The replicating unit is therefore selected by an agent that never plays the game, according to a functional that is not the social one. Selection here is artificial rather than natural, in the breeder’s sense: the copied variant is the one that served the breeder.

The consequences of this decoupling are the subject of this paper. They are not the familiar observation that individual incentives can diverge from collective welfare, which has been understood since Hardin’s account of the commons [4] and is the content of every social dilemma. That divergence is not the point here. The point is structural and sharper: the dynamical system that describes the deployed population is the replicator equation of a game that nobody plays. Its rest points are the equilibria of the principals’ selection game, and the social game enters only as a passive observable. Interventions therefore have very different effects depending on which of the two objects they touch. An intervention on the design pool — safety training, capability evaluations, release gating — changes the set of available strategies. An intervention on the selection functional — liability, disclosure, procurement rules — changes the vector field itself. We show that the first class can be exactly neutral while the second class is not, and that the second class can be non-monotone in a way that makes intermediate regimes worse than none.

Concern about selection pressure over deployed AI systems has been

raised informally [5] and appears as one of seven risk factors in a recent multi-agent risk taxonomy [6], but we are not aware of a formal treatment. Formal analyses of AI races have instead studied the race itself: whether speed dominates safety [7], how regulation and heterogeneity change the prevalence of unsafe development [8, 9], and how commitments and delegation alter outcomes [10, 11, 12]. Those models keep the classical identity: the racing actors are the reproducing actors. We keep their interaction layer and change the selection layer.

Contribution. We formalise deployment-layer selection, embed it in the reduced AI-race game of Fernández Domingos and Han [13], and derive its consequences analytically and numerically. Section 3 specifies the two layers, Section 8 checks the results in finite populations and Section 10 draws out what they imply for governance. The five results are:

1. **Reduction and inertness** (Section 4). The liability pass-through λ and the per-action external harm h enter the dynamics only through the product $L = \lambda h$. The wedge leaves the replicator orbits unchanged exactly when it is column-constant, which in this game happens only at $\lambda = 1$. Every invasion condition is affine in L , giving closed-form critical values $L^* = \delta a / \delta m$.
2. **Solo evaluation is futile** (Section 5). The conditionally antisocial design CAS weakly dominates the unconditionally unsafe design AU at every $L \geq 0$. A filter that rejects designs whose unsafe frequency against a safe environment exceeds ε therefore rejects AU and admits CAS, whose self-play unsafe frequency is 1. The only attractor the filter deletes is one with maximal harm that the surviving pool reproduces, and on a common start measure the long-run unsafe frequency is unchanged.
3. **The safe face is neutral and its barrier is not** (Section 6). Always-safe and conditionally-safe designs are exact payoff twins, so the safe face carries no selection gradient. Its invasion barrier nonetheless varies from $L^* = 42.77$ at the unconditional vertex to $L^* = 0.55$ at the conditional vertex, a factor of 77.7. Protection is therefore governed by a variable that drifts.
4. **Long-run harm is non-monotone in liability** (Section 7). On $L \in (4.27, 42.77)$ the interior rest point of the unsafe face becomes uninvadable by conditional safety, because liability also taxes retaliation. Crossing the lower edge from below raises the basin-averaged unsafe frequency discontinuously from 0 to 0.32.

5. Irreversible diffusion makes the transition hysteretic (Section 9).

With a monotone environmental state that erodes enforcement by a factor θ , the liability at which a safe regime is recovered is $1/(1 - \theta)$ times the liability at which it was lost.

Two limitations should be stated before the results rather than after them. The interaction layer is the deliberately stylised two-player race of [13] with four reduced strategies; it abstracts away many actors, gradual capability, and learning. And the external harm h is a free parameter with no calibration; we therefore report the weight-free unsafe frequency as the primary observable and characterise the entire L -axis rather than committing to a point estimate.

2. Related work

Evolutionary models of AI races. Armstrong et al. [7] introduced a formal race model in which information about rivals changes the willingness to skip safety precautions. Han et al. [8] embedded a race in an evolutionary dynamic and showed that whether regulation is beneficial depends on the speed advantage of unsafe development and the time horizon; Cimpanu et al. [9] extended this to heterogeneous populations. These models establish that competitive pressure alone can sustain unsafe development. They also share a modelling choice we depart from: the racing actor and the reproducing actor coincide, so the population dynamic is driven by the racing payoff itself. Our contribution is orthogonal rather than corrective — we retain their race and ask what changes when the replication decision is made by a third party.

Delegation and artificial agents. Fernández Domingos et al. [11] showed experimentally that delegating a collective-risk decision to an artificial agent increases prosocial behaviour, and Terrucha et al. [12] showed that committing to an imperfect delegate can beat acting directly. That literature treats delegation as a commitment device chosen once by a player inside the game. Here the principal is outside the game and acts repeatedly on a population of designs; the object of interest is not the delegate’s behaviour but the selection pressure the principal exerts on the design distribution.

Behavioural evidence on the race. The interaction layer we adopt is the framed experiment of [13], in which paired participants repeatedly chose between safe and unsafe development under an uncertain horizon. Its central finding is that unsafe choices track the evolving strategic state of the race — the opponent’s previous action and the progress gap — rather than elicited

risk preferences, which motivates the conditional strategies CS and CAS that carry all of our results. Conditional reciprocation of this kind is the dominant behavioural regularity in repeated social dilemmas more broadly [14, 15, 16].

Environmental feedback in evolutionary games... Coupling replicator dynamics to a slow environmental state produces oscillations and bistability [17, 18], and abrupt regime shifts with hysteresis are a well-studied feature of ecosystems with such feedbacks [19]. Existing game–environment models take the environment to be renewable: it recovers when exploitation stops. Deployed capability does not. Our ratchet is the degenerate but practically important case of a state that is monotone by construction, which converts a reversible parameter change into an irreversible regime change.

Liability. The economic analysis of accident law asks when strict liability outperforms negligence and when liability should be combined with regulation [20, 21]. Both analyses are equilibrium statements about a fixed population of injurers. We ask instead what a liability level does to an evolving distribution of designs, and find a qualitatively different phenomenon: a window of intermediate liability that is worse than either bound because it taxes the retaliation on which conditional safety depends. That taxation of retaliation is a second-order free-rider problem [22, 23] in a liability setting.

3. Model

We now specify the two layers. The interaction layer is taken unchanged from the source study [13] so that our results are attributable to the selection layer alone; the selection layer is new.

3.1. Interaction layer: the reduced AI race

Two agents repeatedly choose $a \in \{S, U\}$, safe or unsafe development. Safe development advances the race by $\sigma(S) = 1$ step and unsafe development by $\sigma(U) = 1.5$ steps. The round payoff matrix, rows indexed by the focal action, is

$$\pi = \begin{pmatrix} 1 & 0.6 \\ 2.4 & 2 \end{pmatrix}. \quad (1)$$

The race lasts at least five rounds and each further round ends it with probability 0.2, so the horizon is $T = 5 + G$ with $G \sim \text{Geom}(0.2) - 1$ and $\mathbb{E}[T] = 9$. The agent with greater cumulative progress receives a prize $B = 100$; equal progress splits it. A winning or tying agent suffers a setback with probability

$$q_i(T) = p_r^{\max} \frac{n_i^U(T)}{T}, \quad (2)$$

where $n_i^U(T)$ counts its unsafe choices, and a setback removes its task payoff. We use $p_r^{\max} = 0.6$ throughout and report the other two experimental treatments in [Appendix B](#).

The reduced strategy set is the one inferred from the experiment: AS always safe; AU always unsafe; CS safe in round one, then copying the opponent’s previous action; CAS unsafe in round one, then copying.

Remark 1 (Exact evaluation). All four designs are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is T . We therefore obtain the matchup quantities by exact expectation over the horizon law rather than by the Monte Carlo procedure used in the source study [13]. Exactness matters here because several of our thresholds are ratios of small payoff differences, so sampling noise of order 10^{-2} in the payoff matrix would propagate to the third significant digit of those ratios.

Write $a(i, j)$ for the expected task payoff of design i against j , $m(i, j)$ for its expected number of unsafe actions, and $u(i, j) = \mathbb{E}[n^U/T]$ for its expected unsafe frequency. Table [A.1](#) reports a and Table [A.2](#) reports m ; both are reproduced with the threshold table in [Appendix A](#).

3.2. Deployment layer: three functionals and the wedge

Each unsafe action imposes an external harm $h \geq 0$ on parties outside the race. Let $\lambda \in [0, 1]$ be the fraction of that harm borne by the principal who deployed the design — the liability pass-through. That single parameter splits the interaction into three functionals:

$$\text{task payoff: } a(i, j), \tag{3}$$

$$\text{selection functional: } \pi_P(i, j) = a(i, j) - \lambda h m(i, j), \tag{4}$$

$$\text{social functional: } \pi_S(i, j) = a(i, j) - h m(i, j). \tag{5}$$

Only π_P enters the dynamics. The social functional (5) is an accounting device: since each agent’s harm is attributed to itself, the population mean of π_S equals mean task payoff minus total external harm.

Definition 2 (Selection wedge). The selection wedge is $\Delta(i, j) = \pi_P(i, j) - \pi_S(i, j) = (1 - \lambda)h m(i, j)$.

Figure [1](#) summarises the construction. The wedge is the entire formal content of the separation between the layer at which safety is engineered and the layer at which designs are selected.

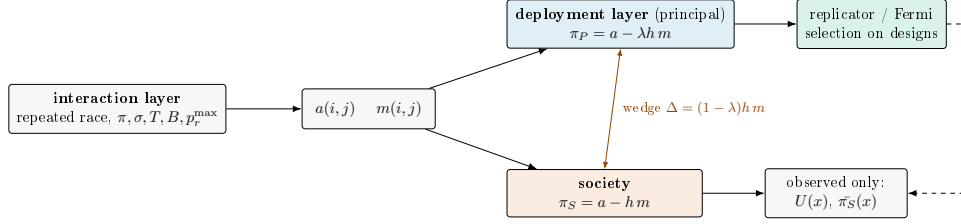


Figure 1: Deployment-layer selection. The interaction layer produces a task payoff a and an externality intensity m . The principal selects designs on π_P , which internalises a fraction λ of the harm; society is scored by π_S , which internalises all of it. Only the upper path feeds back into the population dynamics, so the social functional and the unsafe frequency $U(x)$ are observables of a system that does not optimise them. Their difference is the selection wedge Δ .

3.3. Dynamics

Let $x \in \Delta_n$ be the frequency distribution over the n available designs. In an infinite population the deployment dynamics is the replicator equation of the selection functional,

$$\dot{x}_i = x_i \left[(\pi_P x)_i - x^\top \pi_P x \right], \quad i = 1, \dots, n. \quad (6)$$

In a finite population of size Z we use the pairwise-comparison process with Fermi imitation [24]: a design B adopts A with probability $(1 + \exp[-\beta(f_A - f_B)])^{-1}$, where β is the intensity of selection and f is computed from π_P , with mutation probability μ . We report both the full chain and its small-mutation limit [25]. All computations use EGTtools [26].

The primary observable is the population unsafe frequency

$$U(x) = \sum_{i,j} x_i x_j u(i, j), \quad (7)$$

which requires no welfare weights and is therefore independent of the uncalibrated harm parameter.

4. Reduction, inertness and invasion thresholds

This section establishes the analytical scaffolding: the two-parameter family (λ, h) collapses to one parameter, the wedge is dynamically inert only in a degenerate case, and all invasion conditions are affine.

Proposition 3 (Reduction to the effective liability). *The dynamics (6) and the Fermi process depend on (λ, h) only through the product $L = \lambda h$.*

Proof. Both dynamics depend on π_P only, and by (4) $\pi_P = a - Lm$. $\square$

We call L the *effective liability*, measured in payoff units per unsafe action. It is the only policy variable in what follows; a regulator can move it by raising the pass-through λ at fixed harm, and the same outcome obtains for any (λ, h) on the hyperbola $\lambda h = L$.

Lemma 4 (Column-constant invariance). *Let $A' = A + \mathbf{1}c^\top$ for some $c \in \mathbb{R}^n$. Then A and A' induce the same replicator field on Δ_n and have the same symmetric Nash set.*

Proof. $(A'x)_i = (Ax)_i + c^\top x$ for every i , so all fitnesses shift by the same state-dependent constant and the differences $(A'x)_i - x^\top A'x$ are unchanged; likewise every pairwise payoff comparison at a fixed opponent distribution is unchanged. $\square$

Definition 5. A wedge Δ is *strategically inert* if $\Delta = \mathbf{1}c^\top$, i.e. $\Delta(i, j)$ does not depend on the focal design i .

Proposition 6 (Inertness only at full pass-through). *In the reduced race game, $\Delta = (1 - \lambda)hm$ is strategically inert if and only if $\lambda = 1$ or $h = 0$. Consequently the deployment dynamics is orbit-equivalent to the dynamics of the social functional exactly at full liability pass-through.*

Proof. By Lemma 4 inertness holds iff $m(i, j)$ is independent of i or the prefactor vanishes. Table A.2 gives $m(\text{AS}, \text{AS}) = 0 \neq 9 = m(\text{AU}, \text{AS})$, so m depends on the focal design and inertness requires $(1 - \lambda)h = 0$. $\square$

Proposition 6 is a negative result with a practical reading: there is no cheap partial liability that reproduces the socially correct dynamics. Any $\lambda < 1$ deforms the vector field, and the deformation is proportional to $1 - \lambda$. It does not, however, follow that the deformation is harmful in proportion to its size; Section 7 shows the opposite for a range of L .

Lemma 7 (Affine invasion conditions). *A rare design i in a resident population of j has positive growth rate under (6) if and only if*

$$\delta a - L \delta m > 0, \quad \delta a := a(i, j) - a(j, j), \quad \delta m := m(i, j) - m(j, j). \quad (8)$$

If $\delta m \neq 0$ the critical effective liability is $L^ = \delta a / \delta m$; the invader is favoured for $L < L^*$ when $\delta m > 0$ and for $L > L^*$ when $\delta m < 0$. If $\delta m = 0$ the outcome does not depend on L .*

Proof. The growth rate of a rare i is $\pi_P(i, j) - \pi_P(j, j)$, which by (4) equals $\delta a - L \delta m$. The case distinction follows by solving a linear inequality. $\square$

Applying Lemma 7 to every ordered pair gives Table A.3, which lists all twelve thresholds. Four of them organise the rest of the paper:

$$\begin{array}{ccccc} \underbrace{L_{\text{CS} \rightarrow \text{CAS}}^* = 0.116}_{\text{conditional safety enters}} & < & \underbrace{L_{\text{CAS} \rightarrow \text{CS}}^* = 0.551}_{\text{conditional safety protected}} & & \\ & & & & \\ & < & \underbrace{L^\dagger = 4.274}_{\text{retaliation becomes too costly}} & < & \underbrace{L_{\text{CAS} \rightarrow \text{AS}}^* = 42.765}_{\text{unconditional safety protected}} . \end{array} \quad (9)$$

The spread of (9) over more than two orders of magnitude is the quantitative signature of the results below.

5. Solo evaluation cannot move the equilibrium

Pre-deployment safety evaluation probes a model in isolation and admits it when its behaviour is acceptable. We formalise this as a filter on the design pool and show that it is exactly neutral.

Definition 8 (Solo-evaluation filter). The solo score of design i is $s(i) = u(i, \text{AS})$, its unsafe frequency against a fully safe environment. The filter of strictness ε admits $\Theta_\varepsilon = \{i : s(i) \leq \varepsilon\}$.

In the reduced game, $s(\text{AS}) = s(\text{CS}) = 0$, $s(\text{AU}) = 1$, and CAS defects exactly once before its opponent's safe play locks it into safe play, so

$$s(\text{CAS}) = \mathbb{E}\left[\frac{1}{T}\right] = \sum_{g \geq 0} \frac{0.2 \cdot 0.8^g}{5 + g} = 0.132. \quad (10)$$

Any filter with $\varepsilon \in [0.132, 1)$ therefore admits $\{\text{AS}, \text{CS}, \text{CAS}\}$ and rejects only AU. The next result says that the rejection is dynamically empty.

Theorem 9 (Weak dominance of CAS over AU). *For every $L \geq 0$ and every opponent j , $\pi_P(\text{CAS}, j) \geq \pi_P(\text{AU}, j)$, with strict inequality for $j \in \{\text{AS}, \text{CS}\}$.*

Proof. From Tables A.1 and A.2,

$$\pi_P(\text{CAS}, j) - \pi_P(\text{AU}, j) = \begin{cases} 53.125 + 8L, & j = \text{AS}, \\ 14.271 + 4.222L, & j = \text{CS}, \\ 0, & j \in \{\text{AU}, \text{CAS}\}, \end{cases} \quad (11)$$

each of which is non-negative for $L \geq 0$ and strictly positive in the first two cases. Against AU and CAS the two designs generate identical action paths from round two onwards and identical unsafe counts, so the difference vanishes identically. $\square$

The mechanism behind Theorem 9 is transparent: AU and CAS behave the same way against an unsafe environment, but CAS stops defecting against a safe one, which earns it more and costs it less. Unconditional aggression is dominated by conditional aggression at every liability level.

Corollary 10 (Filter futility). *Let $\Theta = \{AS, AU, CS, CAS\}$ and $\varepsilon \in [0.132, 1)$, so that $\Theta_\varepsilon = \{AS, CS, CAS\}$. Then, for every $L \geq 0$:*

- (i) *along every interior solution of (6) the ratio x_{AU}/x_{CAS} is non-increasing, and strictly decreasing whenever $x_{AS} + x_{CS} > 0$;*
- (ii) *any attractor of Θ that is removed by the filter carries the maximal unsafe frequency $U = 1$; the filter therefore cannot delete an attractor with lower harm than one it leaves behind;*
- (iii) *under a start measure that is common to both pools — initial conditions drawn on Θ_ε with a fixed small admixture of AU — the basin-averaged long-run unsafe frequencies of Θ and Θ_ε agree to within 1.2×10^{-3} across the whole L -axis.*

Proof. For (i), (11) gives $\frac{d}{dt} \log(x_{AU}/x_{CAS}) = \pi_P(AU, x) - \pi_P(CAS, x) = -[(53.125 + 8L)x_{AS} + (14.271 + 4.222L)x_{CS}] \leq 0$. For (ii), by (i) an ω -limit with $x_{AU} > 0$ must satisfy $x_{AS} = x_{CS} = 0$ and hence lie in the $\{AU, CAS\}$ edge, on which the two designs are payoff-identical, every point is a rest point, and $u \equiv 1$. Part (iii) is numerical and reported in Figure 2A. $\square$

Part (ii) is worth stating plainly, because it is weaker than one might hope and that weakness is itself informative. Filtering AU does delete an attractor: for $L < L_{AS \rightarrow AU}^* = 2.422$ an AU-dominated population is neutrally stable, and removing AU removes it. But that attractor has $U = 1$, exactly the unsafe frequency of the CAS vertex that the filter leaves in place whenever $L < L_{CS \rightarrow CAS}^* = 0.116$. The filter changes which design occupies the all-unsafe state, not whether the state exists, and it is reachable only from populations already dominated by AU.

Figure 2 makes the comparison on a common start measure, so that the two pools describe the same ecosystem with and without a small admixture of the filtered design rather than two different sampling measures. On that comparison the curves coincide over the whole L -axis. Panel B shows why. A solo evaluation sees $s(CAS) = 0.132$ and cannot distinguish CAS from a

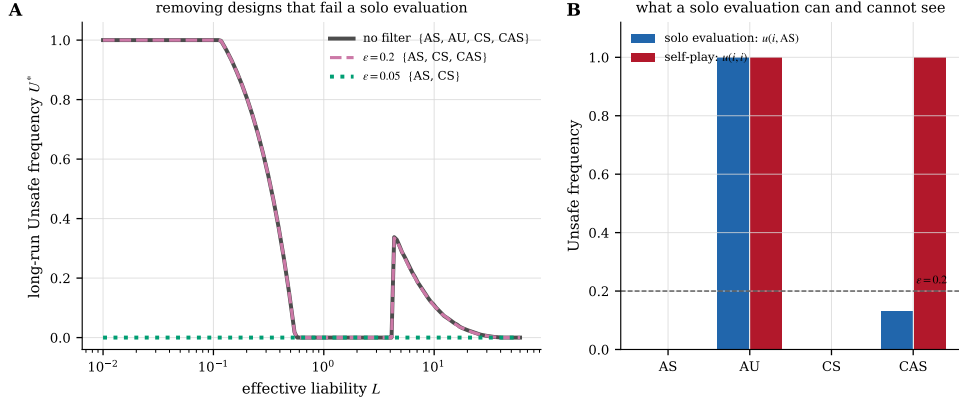


Figure 2: Solo evaluation filters are dynamically empty. **(A)** Long-run unsafe frequency of the unfiltered pool $\{AS, AU, CS, CAS\}$ and of the pools admitted at $\varepsilon = 0.2$ and $\varepsilon = 0.05$, as a function of the effective liability. Initial conditions are drawn on the $\{AS, CS, CAS\}$ simplex for every pool, with AU seeded at a fixed 5% where it is present, so that the three curves describe the same ecosystem. The first two coincide everywhere, as Corollary 10(iii) requires. **(B)** What the evaluation sees. Blue: unsafe frequency against a safe environment, the quantity the filter measures. Red: unsafe frequency in self-play, the quantity that determines equilibrium harm. The two orderings disagree precisely for CAS.

safe design at any threshold that admits reciprocation at all, while CAS in self-play produces $u = 1$ — the same equilibrium harm as the design the filter did remove.

A strict filter with $\varepsilon < 0.132$ does remove CAS, leaving the pool $\{AS, CS\}$ on which $U^* = 0$ for every L . Two caveats make this less reassuring than it looks. First, that pool is exactly the neutral safe face of Section 6, so its protection lasts only as long as no CAS-like design re-enters, and Theorem 12 bounds how much liability is needed when one does. Second, the strict threshold is only achievable because the probe environment in (10) is fully safe: measured against an unsafe opponent the conditionally safe design scores $u(CS, CAS) = 0.461$ and would be rejected by the same rule. A filter strict enough to exclude conditional aggression in a mixed environment also excludes conditional safety, which Theorem 12 identifies as the cheapest design to protect.

6. A neutral safe face with a drifting barrier

Proposition 11 (Payoff twins). *Restricted to $\{AS, CS\}$, the selection functional is constant: $\pi_P(i, j) = a(AS, AS) = 59$ for all $i, j \in \{AS, CS\}$ and all*

L . Consequently the AS-CS edge is a set of rest points of (6), and in the finite-population process it evolves by neutral drift.

Proof. A conditionally safe design plays safe in round one and then copies a safe opponent, so all four matchups within $\{\text{AS}, \text{CS}\}$ realise the all-safe path. Hence $a \equiv 59$, $m \equiv 0$ and $\pi_P \equiv 59$ independently of L . $\square$

Neutrality does not make the composition irrelevant, because AS and CS differ against an invader.

Theorem 12 (Composition-dependent invasion barrier). *Let the resident population be the mixture $(1-x)\text{AS} + x\text{CS}$ on the safe face. A rare CAS invader has positive growth rate if and only if $L < L^*(x)$, where*

$$L^*(x) = \frac{(1-x)\delta a_{\text{AS}} + x\delta a_{\text{CS}}}{(1-x)\delta m_{\text{AS}} + x\delta m_{\text{CS}}}, \quad \begin{array}{ll} \delta a_{\text{AS}} = 42.765, & \delta m_{\text{AS}} = 1, \\ \delta a_{\text{CS}} = 2.631, & \delta m_{\text{CS}} = 4.778. \end{array} \quad (12)$$

L^* is strictly decreasing on $[0, 1]$, with $L^*(0) = 42.765$ and $L^*(1) = 0.551$, a ratio of 77.7.

Proof. By Proposition 11 the resident fitness is 59 for every x , so the invader's growth rate is $(1-x)[\pi_P(\text{CAS}, \text{AS}) - 59] + x[\pi_P(\text{CAS}, \text{CS}) - 59] = (1-x)(\delta a_{\text{AS}} - L\delta m_{\text{AS}}) + x(\delta a_{\text{CS}} - L\delta m_{\text{CS}})$, which is positive exactly when $L < L^*(x)$. Differentiating the ratio of the two affine functions, the sign of $\frac{d}{dx}L^*$ is that of $\delta a_{\text{CS}}\delta m_{\text{AS}} - \delta a_{\text{AS}}\delta m_{\text{CS}} = 2.631 - 204.34 < 0$. $\square$

Inverting (12) gives the critical share of conditional safety $x^*(L)$ that a given liability level requires: $x^*(1) = 0.951$, $x^*(2) = 0.855$, $x^*(5) = 0.640$, $x^*(20) = 0.197$. A population that is 90% unconditionally safe is invadable at any $L < 28.1$, whereas the same population with the shares reversed is already protected at $L = 1.6$.

The combination of Proposition 11 and Theorem 12 is the substantive point. The variable that determines whether a safe population can be invaded carries no selection gradient of its own, so it is free to drift. Figure 3C shows the stationary distribution of the neutral edge in a finite population: for small mutation the mass concentrates at the two vertices, so the population spends long stretches near pure AS, which is the maximally vulnerable composition. Protection obtained by mandating unconditional safety is therefore not durable — not because the mandate is violated, but because the direction it protects against is selectively invisible.

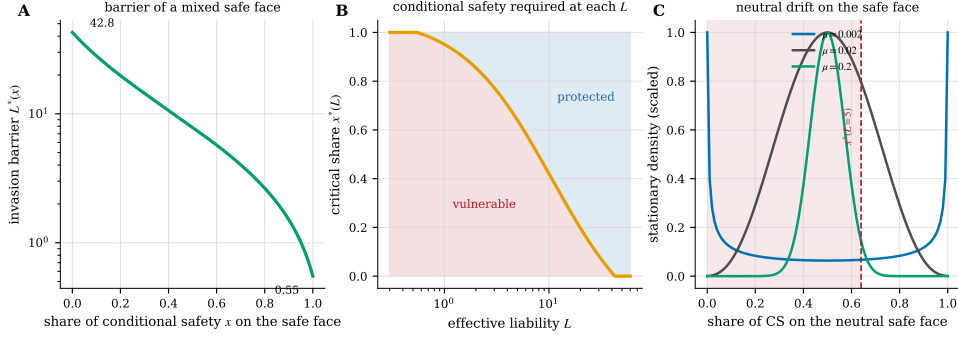


Figure 3: The safe face is neutral but not uniform. **(A)** Invasion barrier $L^*(x)$ of Theorem 12, log scale. **(B)** The critical share of conditional safety $x^*(L)$ separating vulnerable from protected compositions. **(C)** Stationary density of the neutral AS–CS edge in a finite population ($Z = 100$, $\beta = 0.05$) for three mutation rates, with the vulnerability threshold at $L = 5$ marked. Small mutation concentrates the mass at the vertices, half of it on the vulnerable side.

7. Bistability and the liability valley

The results so far might suggest that harm decreases with liability. It does not. This section identifies a window of intermediate liability on which the system is bistable and the basin-averaged harm jumps upward.

Proposition 13 (Interior rest point of the unsafe face). *On the $\{AS, CAS\}$ face the replicator field has an interior rest point whenever $\alpha := \pi_P(CAS, AS) - \pi_P(AS, AS)$ and $\gamma := \pi_P(AS, CAS) - \pi_P(CAS, CAS)$ share a sign, located at $p^* = \alpha/(\alpha + \gamma)$ in the CAS direction, and it attracts the face when both are positive. With the values of Table A.1,*

$$p^*(L) = \frac{42.765 - L}{24.165 + 8L}, \quad L \in (L_{AS \rightarrow CAS}^*, L_{CAS \rightarrow AS}^*) = (2.067, 42.765). \quad (13)$$

Proof. Direct computation of the two-strategy replicator rest point, followed by substitution of $\alpha = 42.765 - L$ and $\gamma = 9L - 18.6$. $\square$

Theorem 14 (Mixture-free invasion criterion). *At any interior rest point of the $\{AS, CAS\}$ face, the growth rate of a rare CS is*

$$p^* \left[(a(CS, CAS) - a(AS, CAS)) - L(m(CS, CAS) - m(AS, CAS)) \right], \quad (14)$$

whose sign does not depend on p^* . Hence CS invades the unsafe coexistence if and only if

$$L < L^\dagger := \frac{a(\text{CS}, \text{CAS}) - a(\text{AS}, \text{CAS})}{m(\text{CS}, \text{CAS}) - m(\text{AS}, \text{CAS})} = \frac{26.644 - 8.600}{4.222 - 0} = 4.274. \quad (15)$$

Proof. At the rest point AS and CAS have equal fitness f , so the growth rate of CS is $\pi_P(\text{CS}, x) - f = \pi_P(\text{CS}, x) - \pi_P(\text{AS}, x)$. Expanding and using $\pi_P(\text{CS}, \text{AS}) = \pi_P(\text{AS}, \text{AS})$ from Proposition 11, the terms proportional to $1 - p^*$ cancel and (14) remains. $\square$

The interpretation of $L^\dagger$ is the crux of the paper. A conditionally safe design defends a population by retaliating: it matches unsafe play, which is what makes exploiting it unprofitable. Retaliation is itself unsafe behaviour, so liability taxes it. Above $L^\dagger$ the tax on retaliation exceeds the tax the retaliation imposes on the aggressor, the guard loses its advantage, and the unsafe coexistence becomes uninvadable.

Corollary 15 (Liability valley). *For $L \in (L^\dagger, L_{\text{CAS} \rightarrow \text{AS}}^*) = (4.274, 42.765)$, with $L^\dagger$ from (15), the dynamics on $\{\text{AS}, \text{CS}, \text{CAS}\}$ has two attractors: the neutral safe face with $U = 0$, and the interior rest point (13) of Proposition 13 with $U = p^*(1 - p^*)u(\text{CAS}, \text{AS}) + (p^*)^2 > 0$. At the lower edge the unsafe attractor carries $U = 0.465$; weighting the two attractors by the volume of their basins, the long-run unsafe frequency is discontinuous at $L^\dagger$, rising from 0 to 0.32 as L crosses it from below.*

Corollary 15 is the sharpest policy statement available from the model: over an interval spanning a factor of ten in L , more liability means more harm. A regulator moving L upward from a protected regime passes through a range in which the protection it was relying on disappears. It does not recover that protection until L exceeds 42.765, a level at which the external harm of a single unsafe development step is worth more than 42% of the prize for winning the race.

Figure 4 assembles the analytic branches and the two numerical dynamics. The agreement is exact where it should be: the basin-averaged replicator tracks the analytic CS–CAS branch on $(0.116, 0.551)$ and the analytic AS–CAS branch inside the valley, lying slightly below it because a fraction of initial conditions still reaches the safe attractor. Figure 5 shows the corresponding phase portraits.

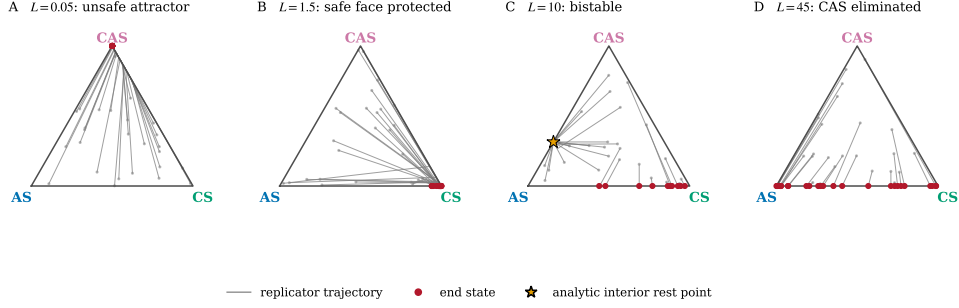


Figure 5: Replicator phase portraits on the $\{AS, CS, CAS\}$ simplex at four liability levels. Trajectories start from random interior points; the star marks the analytic interior rest point of Proposition 13. In the valley (C) the flow separates into two basins; at high liability (D) every trajectory reaches the neutral safe face, where it stops at a history-dependent composition.

9. An eco-evolutionary ratchet

Deployed capability does not disappear. We couple the deployment dynamics to a state $z \in [0, 1]$ measuring the stock of diffused unsafe capability, which enters through enforcement: as a capability proliferates, attributing harm to a particular principal becomes harder, so the liability that actually reaches a principal decays. The coupled system is

$$\dot{x} = \text{Rep}(x; a - L_{\text{eff}}(z)m), \quad \dot{z} = \varepsilon U(x)(1 - z), \quad L_{\text{eff}}(z) = L(1 - \theta z), \quad (16)$$

with $\varepsilon \ll 1$ separating the time scales and $\theta \in [0, 1)$ the maximal erosion. The construction differs from renewable game-environment models [17, 18] in that $\dot{z} \geq 0$ identically: the environmental state is a ratchet.

Theorem 16 (Hysteresis width). *Assume the safe attractor has $U = 0$ exactly, as in Proposition 11. Consider a quasi-static sweep of L that starts in the protected regime. The protected regime is lost at $L_{\downarrow} = L_{CAS \rightarrow CS}^*$, after which $z \rightarrow 1$; it is recovered on the return sweep only at*

$$L_{\uparrow} = \frac{L_{CAS \rightarrow CS}^*}{1 - \theta}, \quad \text{so} \quad \frac{L_{\uparrow}}{L_{\downarrow}} = \frac{1}{1 - \theta}. \quad (17)$$

Proof. On the protected branch $U(x) = 0$, so the second equation of (16) gives $\dot{z} = 0$ and $L_{\text{eff}} = L$; the branch is lost exactly when CAS can invade the resident safe face, i.e. at $L = L_{CAS \rightarrow CS}^*$ by Lemma 7 and (8). On the unsafe branch $U > 0$, so $z \rightarrow 1$ and $L_{\text{eff}} \rightarrow L(1 - \theta)$; recovery requires $L_{\text{eff}} > L_{CAS \rightarrow CS}^*$, which gives (17). $\square$

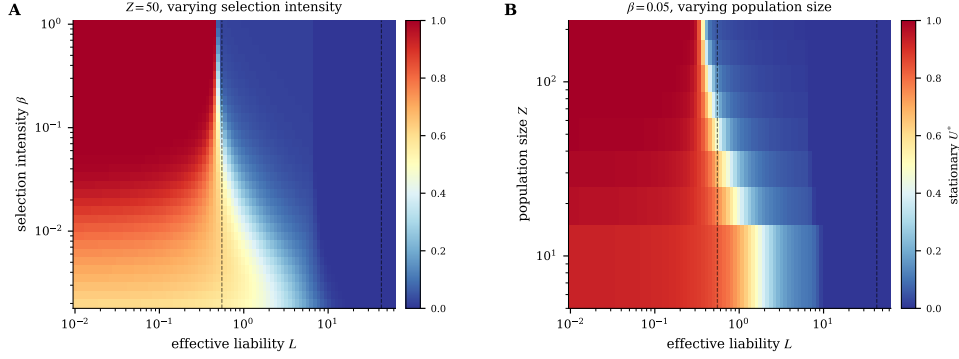


Figure 6: Stationary unsafe frequency of the finite-population pairwise comparison process on $\{\text{AS}, \text{CS}, \text{CAS}\}$ in the small-mutation limit. (A) Varying selection intensity at $Z = 50$. (B) Varying population size at $\beta = 0.05$. Dashed lines mark $L_{\text{CAS} \rightarrow \text{CS}}^*$ and $L_{\text{CAS} \rightarrow \text{AS}}^*$.

Figure 7 confirms the prediction numerically at $\theta = 0.9$: the descending branch loses protection between $L = 0.62$ and $L = 0.54$, bracketing $L_{\text{CAS} \rightarrow \text{CS}}^* = 0.551$, and the ascending branch recovers it at $L = 5.55$, against the predicted 5.51. The loop width is the predicted factor of ten.

The practical content is a timing asymmetry. Because $L_{\uparrow}$ exceeds $L_{\downarrow}$ by $1/(1 - \theta)$, the cost of preventing a transition is smaller than the cost of reversing one by exactly the factor by which enforcement erodes, and the required overshoot is known in closed form once θ is estimated.

10. Discussion

What the model says and does not say. The results are statements about a specific stylised race with four deterministic designs. They are not predictions about frontier AI systems, and the external harm h has no calibration, which is why every result is stated on the whole L -axis. What does transfer is the structure. Whenever the entity that selects a design is not the entity that plays the game, and internalises only part of the harm, the population dynamics is the replicator equation of the selector’s functional. The four mechanisms identified here — dominance of conditional over unconditional aggression, neutrality of the safe face, taxation of retaliation, and irreversibility — then follow from qualitative features of the payoff structure rather than from the particular numbers. The first requires only that a conditional design imitate the unconditional one against unsafe opponents. The second requires only that safe designs be behaviourally indistinguishable against safe opponents. The third requires only that retaliation be classified

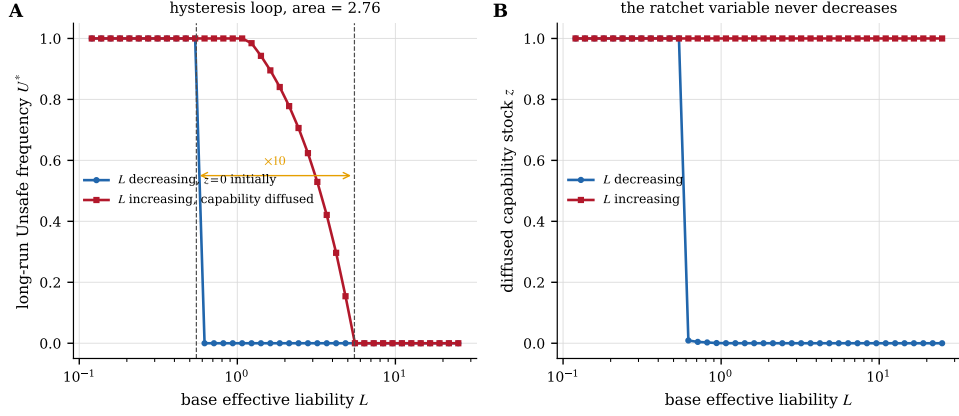


Figure 7: Irreversible capability diffusion produces hysteresis. (A) Quasi-static sweep of the base effective liability, decreasing from a protected state and then increasing again. (B) The ratchet variable, which never decreases. At $\theta = 0.9$ the recovery threshold is ten times the loss threshold, as Theorem 16 predicts.

as harmful by the liability rule.

Evaluations measure the wrong object.. Corollary 10 says that a filter defined on solo behaviour removes a weakly dominated design. That statement is a formal counterpart of a known practical worry, that a model may behave acceptably when probed and unacceptably in deployment [27, 6]. The mechanism here requires no deception: CAS is not hiding anything. It is simply conditional, and a solo probe cannot elicit the condition. The corrective suggested by the model is not stricter thresholds but a different measurement: score designs by $u(i, i)$, the self-play frequency, or more generally against the environment the design will actually meet. Panel B of Figure 2 is the whole argument — the two orderings disagree on exactly one design, and it is the one that governs the equilibrium.

Do not mandate unconditional safety.. Theorem 12 inverts a natural intuition. Unconditional safety looks like the strictest possible design, and it is the most fragile: it has the highest invasion barrier requirement of any safe composition, by a factor of 77.7. Conditional safety is cheap to protect because it makes exploiting it unprofitable. Because the safe face is neutral, a policy that fixes the composition is not self-sustaining; it must be maintained against drift.

Liability has a valley, and it is wide.. Corollary 15 is the result most likely to matter outside the model. Liability instruments are usually assumed to

be monotone: more internalisation, less harm. Here an entire decade of L is worse than the lower end of the range, because the instrument does not distinguish unsafe behaviour that initiates from unsafe behaviour that retaliates. Theorem 14 locates the boundary exactly. The distinction between initiating and retaliating unsafe behaviour is available in principle — it is a matter of conditioning the rule on the counterparty’s prior action, which is exactly the column-constant structure that Proposition 6 identifies as inert. A liability rule assessed on the counterparty’s behaviour rather than one’s own would not deform the dynamics at all. Designing an implementable approximation of that rule is the natural next question.

Limitations. Four are material. The interaction layer has two players and a fixed pool of four designs, so entry of new designs is modelled only as mutation. The principals are homogeneous and passive: they select on π_P but do not strategise about each other, which rules out the enforcement-avoidance behaviour that a richer model would include. The ratchet channel is one of several plausible couplings; erosion of harm attribution is assumed rather than derived. Finally, the analysis is deterministic in h and λ , whereas real liability arrives with probability and delay — an expected-value reading of L is a first approximation only.

11. Conclusion

Separating the payoff that an AI design earns from the payoff that decides whether it is replicated changes the dynamical system, not merely its parameters. Working through that separation in the reduced AI race yields four results that are invisible when the two payoffs are identified. Pre-deployment solo evaluation cannot change the equilibrium, because it removes a weakly dominated design. The invasion barrier protecting a safe population is controlled by a neutrally drifting composition, and varies by a factor of 77.7 along it. Long-run harm is non-monotone in liability, with a bistable window spanning $L \in (4.27, 42.77)$. And irreversible capability diffusion makes the loss of a protected regime cost $1/(1 - \theta)$ times less than its recovery.

Each result points at the same target. Governing an ecosystem of deployed agents by acting on the designs — filtering them, training them, mandating them — addresses the layer that does not carry the selection gradient. Acting on the functional that principals select by does carry it, but the instrument is not monotone, and the range in which it backfires is precisely the range in which it is strong enough to punish retaliation and too weak to protect unconditional safety. Locating that range for a given deployment

setting seems to us more useful than arguing about whether liability should be strengthened.

Two extensions follow directly. Estimating the payoff and harm matrices of real agent designs empirically, rather than from a stylised race, would place an actual deployment ecosystem on the L -axis. And enriching the principals — letting them anticipate liability, compete, or share it — turns the selection functional itself into a strategic object, which is the natural second layer of the problem.

Code and data availability

All code, tables and figures are reproducible from a public repository released under the MIT licence, <https://github.com/trungkiet2005/deployment-layer-selection>. Evolutionary computations use EGT-tools [26].

Appendix A. Matchup tables and thresholds

Table A.1: Task payoffs $a(i, j)$ of the reduced race game at $p_r^{\max} = 0.6$, computed exactly over the horizon distribution. Rows index the focal design.

	AS	AU	CS	CAS
AS	59.000	5.400	59.000	8.600
AU	48.640	27.200	47.360	27.200
CS	59.000	16.600	59.000	26.644
CAS	101.765	27.200	61.631	27.200

Table A.2: Expected number of Unsafe actions $m(i, j)$ of the focal design at $p_r^{\max} = 0.6$.

	AS	AU	CS	CAS
AS	0.000	0.000	0.000	0.000
AU	9.000	9.000	9.000	9.000
CS	0.000	8.000	0.000	4.222
CAS	1.000	9.000	4.778	9.000

Table A.3: Closed-form critical effective liabilities $L^* = \delta a / \delta m$ for every ordered invasion at $p_r^{\max} = 0.6$. The last column states when the invader has a positive growth rate.

Invader	Resident	δa	δm	L^*	invades
AS	AU	-21.800	-9.000	2.422	$L > L^*$
AS	CS	0.000	0.000	–	–
AS	CAS	-18.600	-9.000	2.067	$L > L^*$
AU	AS	-10.360	9.000	-1.151	$L < L^*$
AU	CS	-11.640	9.000	-1.293	$L < L^*$
AU	CAS	0.000	0.000	–	–
CS	AS	0.000	0.000	–	–
CS	AU	-10.600	-1.000	10.600	$L > L^*$
CS	CAS	-0.556	-4.778	0.116	$L > L^*$
CAS	AS	42.765	1.000	42.765	$L < L^*$
CAS	AU	0.000	0.000	–	–
CAS	CS	2.631	4.778	0.551	$L < L^*$

Appendix B. The other risk treatments

The source experiment manipulates the maximum private setback risk $p_r^{\max} \in \{0.1, 0.6, 0.9\}$ in (2). The unsafe counts $m(i, j)$ are identical across treatments, because $p_r^{\max}$ scales the realised payoff but leaves the stage matrix (1) and hence the action paths untouched; only $a(i, j)$ changes. Evaluating the interaction game without liability ($L = 0$) reproduces the qualitative ordering reported in [13]: the neutrally stable set is {AU, CAS} at $p_r^{\max} = 0.1$ and 0.6, and {CS} at $p_r^{\max} = 0.9$, so high private risk favours conditional safety even before any external liability is applied. Our analysis uses $p_r^{\max} = 0.6$, the intermediate treatment, because it is the one in which the conditional designs are both present and neither is trivially dominant.

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